

PRODUCT MANUAL

Operation & Redress Manual - ACT-ST-10 Wireline Pressure Setting Tool (#10)

Document MAN-ST-10 | Revision B | 2026-03-09

Product Family	ACT-ST-10
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Distribution	Field operations, service centers, licensed distributors

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Product Revision History (from engineering records)

Rev	Date	Description of Change	By
A	2021-07-14	Initial release	J. Martinez, P.E.
B	2026-03-02	Firing head o-rings Viton to HNBR per ECO-2026-008	A. Gutierrez

1. GENERAL

The ACT-ST-10 converts #10 power charge gas pressure into a controlled 7.5 in. stroke with up to 60,000 lbf output. An oil-filled metering section (410 mL of 10W hydraulic oil through a 0.028 in. bleed orifice) slows the stroke to protect the element of the tool being set. Rev B tools use HNBR o-rings throughout per ECO-2026-008 for service from -20 degF to 300 degF.

2. SAFETY

WARNING: EXPLOSIVE DEVICE. Assembly, arming and firing are restricted to licensed personnel per API RP 67 and local regulation. Never transport an armed tool.

WARNING: After a misfire, wait the full 60-minute stand-off period before rig-down. Treat the tool as armed until the firing head is removed.

3. OPERATION

Step	Action
1	Verify redress record F-82 current (tool must be redressed after every firing).
2	Check oil fill: remove upper 1/4 NPT fill plug, oil level to bottom thread. Top up with 10W.
3	Install power charge and igniter per charge manufacturer instructions.
4	Make up adapter kit for the tool being set (AK-45, AK-55, FP-AK-55); torque per TS-200.
5	Function limits: max hydrostatic 15,000 psi, max temperature 300 degF, max deviation 90 deg.
6	After firing and POOH: bleed residual gas at the bleed port BEFORE breaking any connection.

4. REDRESS (AFTER EVERY FIRING)

Step	Action
1	Bleed gas; disassemble firing head, chambers, and oil section on a clean bench.
2	Discard all o-rings; clean carbon from chambers with brass brush and solvent.
3	Inspect bleed orifice (0.028 in.): replace if eroded > 0.031 in. gauge pin.
4	MPI pistons and output rod annually or every 25 firings, whichever first.
5	Install kit ACT-ST-10-RK (HNBR 90A o-rings, 3 shear pins); lube with silicone-free grease.
6	Refill 410 mL 10W oil; stroke tool by hand to purge air; verify smooth travel.
7	Replace 3 release shear pins ACT-ST-10-09; stamp redress record F-82.

NOTE: O-ring change per ECO-2026-008: Rev A tools carried Viton -116 o-rings in the firing head; these are superseded by HNBR 90A (ACT-ORG-116-H90) for improved low-temperature sealing. Use up of Viton stock is NOT authorized below 40 degF ambient.

5. STORAGE

Store fully redressed, oil-filled, with thread protectors, horizontal. Re-verify oil level if stored more than 6 months.

Document Approval

Role	Name	Signature	Date
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Prepared	T. Okafor	<i>/signed electronically/</i>	2026-03-09
Quality Review	S. Nguyen	<i>/signed electronically/</i>	2026-03-09
Approved	J. Martinez, P.E.	<i>/signed electronically/</i>	2026-03-09